

Kapture Finance

AI Collections Voicebot

Detailed High-Level Design & Delivery Specification

Voice AI • Vapi • FastAPI • Secure Collections Flow
Submission-ready engineering document

Scenario	Outbound overdue EMI collections
Demo customer	Rahul Sharma - Personal Loan - INR 8,499 - 12 DPD
Assistant	Maya - Kapture Finance
Implementation	Vapi + GPT-4.1 + Soniox STT + Vapi Elliot voice + FastAPI tools

1. Executive Summary

This design specifies a working outbound Voice AI collections agent for Kapture Finance. The design prioritizes privacy, deterministic authentication gating, reliable tool execution, compliant conversational behavior, and operational observability. The demo implementation uses Vapi for the real-time voice pipeline and a FastAPI service for mocked business actions. The solution is intentionally lightweight but structured so that the same interfaces can be replaced by production systems later.

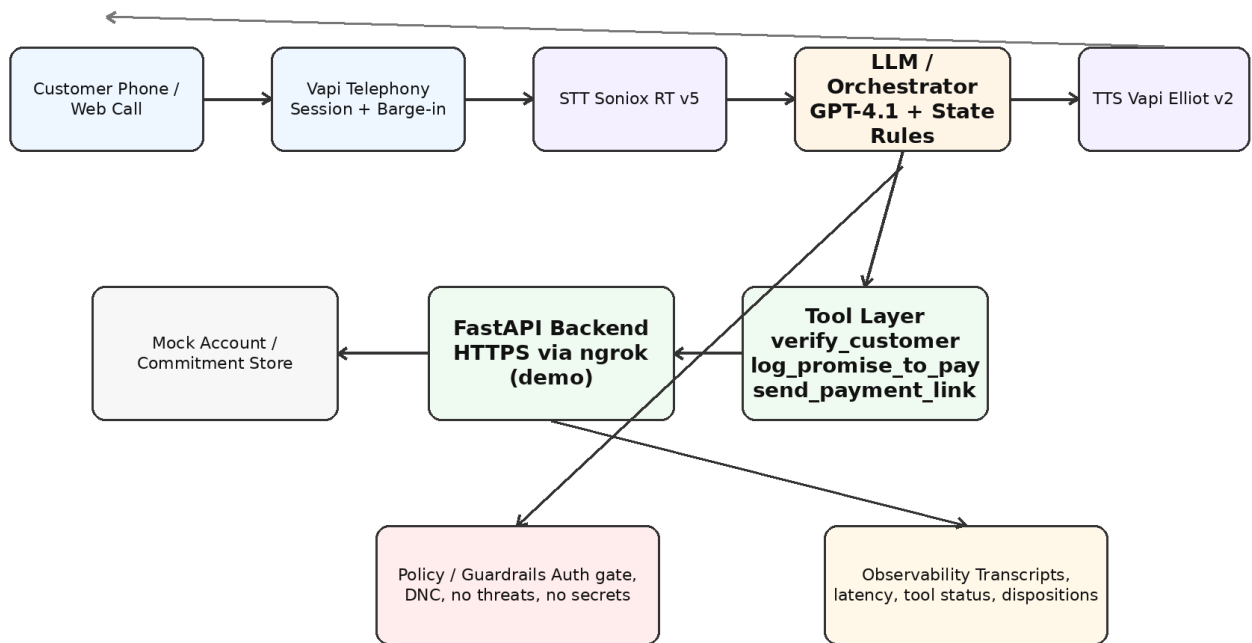
- Primary objective: authenticate the intended customer before disclosing any debt information.
- Business objective: improve containment and promise-to-pay conversion on routine collections calls while routing exceptions safely.
- Engineering objective: keep the conversation responsive, auditable, tool-driven, and recoverable when services fail.

2. Scope, Assumptions, and Non-Goals

Area	Decision / Assumption
Scope	Single outbound collections voicebot, English-first, demo customer Rahul Sharma.
Authentication	Four-digit demo verification code; production would use a stronger verified-data challenge.
Payment	No real money movement. The payment-link endpoint generates a demo URL only.
Data store	In-memory/mock semantics for assignment. Production would use CRM/loan servicing systems.
Calling policy	Calls are assumed to be initiated only during client-approved/permitted hours.
Non-goal	No production-grade PCI flows, no real OTP collection, no automatic legal escalation.
Bonus-ready	Hindi switching, SMS/WhatsApp delivery, evaluation harness, disposition analytics.

3. Architecture & Voice Pipeline

Kapture Finance Voice AI - Reference Architecture



Target conversational latency: ~1.5 s nominal end-to-end; tool calls should typically complete within ~0.3-0.8 s in demo.

The voice pipeline is split into real-time media processing and business-tool execution. Vapi manages call media, transcription, LLM orchestration, and speech synthesis. The LLM is allowed to request tools, but debt disclosure is constrained by an authentication state rule. Business actions are performed through HTTPS APIs so that they can be independently audited and replaced with production systems.

3.1 Component Responsibilities

Component	Responsibility	Failure behavior
Telephony / Web Call	Connect caller, stream audio, handle interruption/barge-in.	Retry call/session according to platform policy; log end reason.
STT - Soniox RT v5	Convert caller audio to text with low latency.	Ask concise clarification on low-confidence/malformed inputs.
LLM - GPT-4.1	Intent recognition, dialog generation, tool selection, state progression.	Do not invent tool results; remain in safe state if uncertain.
TTS - Vapi Elliot v2	Speak short, natural responses.	Fallback voice if configured; otherwise close gracefully.
FastAPI Tools	Verification, PTP logging, payment-link generation.	Return explicit success/failure; LLM confirms only successful actions.
Datastore / CRM adapter	Source of customer/account data and recorded dispositions.	No data -> no disclosure/action; escalate or close safely.
Observability	Capture latency, tool outcomes,	Alert on spikes in failures,

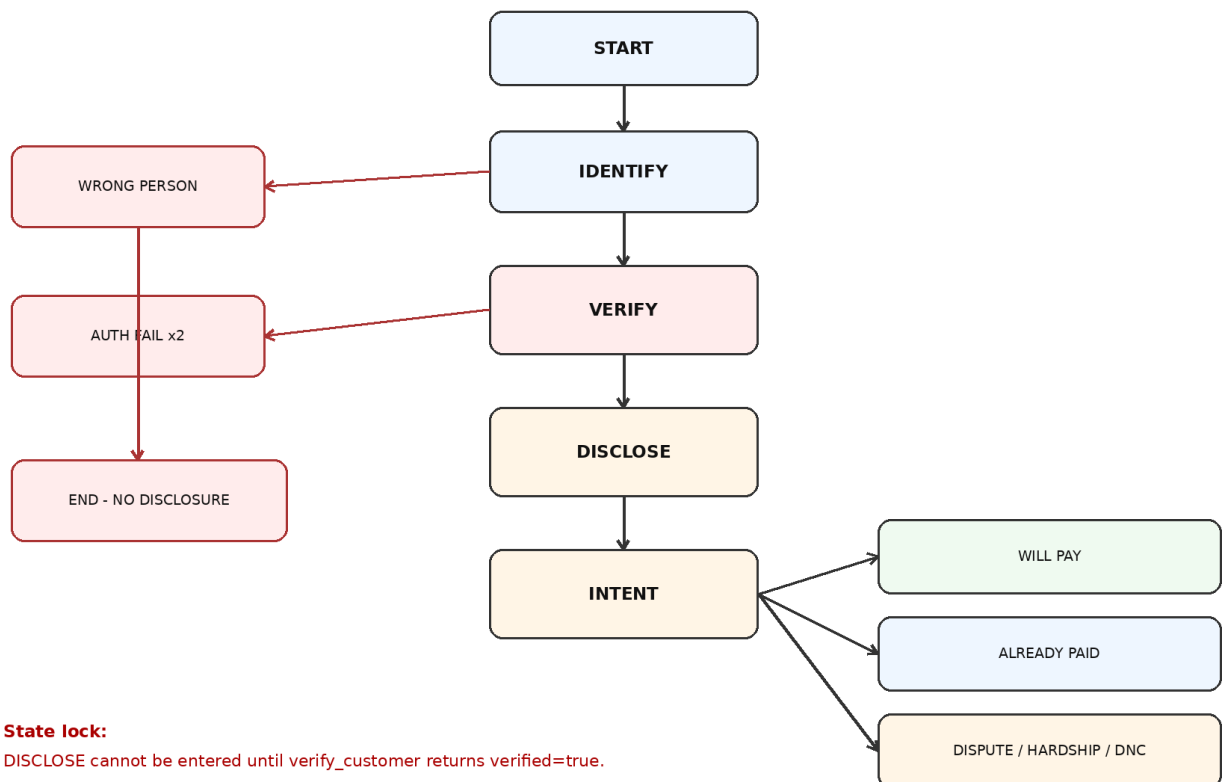
Component	Responsibility	Failure behavior
	state transitions, end reasons.	latency, or privacy violations.

3.2 Latency Budget

Hop	Target	Notes
Audio capture / telephony	80-150 ms	Network dependent.
STT partial/final	250-450 ms	Streaming recognition preferred.
LLM first token / decision	450-750 ms	Prompt kept structured; tool calls add extra time.
Tool/API call	200-800 ms	Demo via ngrok may be slower than production.
TTS first audio	250-450 ms	Short responses reduce perceived delay.
Nominal end-to-end	~1.2-2.0 s	Target for natural turn-taking; current preset ~1.53 s baseline.

4. Conversation State Machine

Conversation State Machine - Authentication Gate is Mandatory



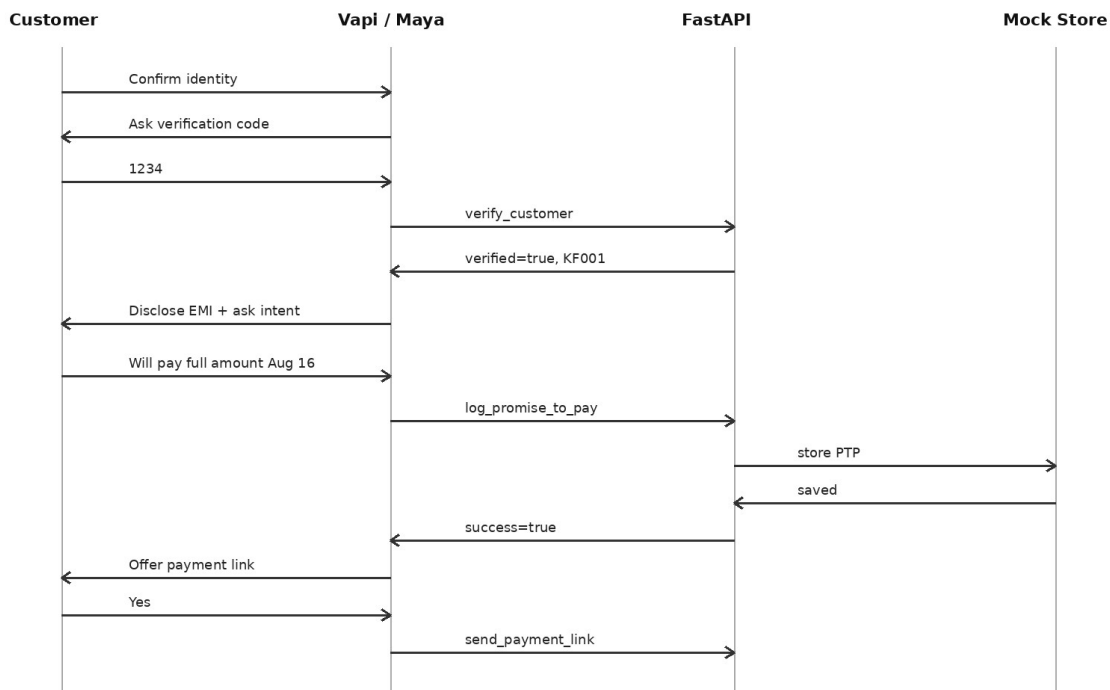
The key control is the transition lock between VERIFY and DISCLOSE. The assistant may not enter the debt-disclosure state merely because the caller claims to be Rahul. The `verify_customer` tool must return `verified=true`. This makes the authentication gate explicit and testable rather than relying only on conversational discretion.

4.1 State Definitions

State	Entry condition	Allowed actions	Exit
IDENTIFY	Call connected	Introduce self/company; ask for Rahul only.	Rahul / wrong person / no response
VERIFY	Caller claims to be Rahul	Ask approved verification challenge; call verify_customer.	verified=true, retry, or end
DISCLOSE	verified=true only	State overdue account purpose/amount; ask intent.	Intent classified
NEGOTIATE / RESOLVE	Intent known	PTP, hardship response, dispute acknowledgement, callback/DNC handling.	Action success or escalation
CLOSE	Outcome determined	Confirm only successful actions; polite close.	Call ended + disposition logged

5. Successful Promise-to-Pay Sequence

Successful Promise-to-Pay Sequence



This sequence demonstrates the core happy path used in the live demo: identity confirmation, tool-based verification, debt disclosure, capture of a promised date, recording of the PTP, optional payment-link generation, and a truthful close.

6. Intent & Entity Model

Intent	Example utterance	Entities extracted	Action
will_pay	I will pay the full amount on August 16.	amount, PTP date	log_promise_to_pay; optionally send_payment_link
cannot_pay / hardship	I lost my job and cannot pay now.	reason if volunteered, callback preference	Empathetic response; offer appropriate follow-up; no invented plan.
already_paid	I paid yesterday.	payment date/channel if volunteered	Acknowledge; stop pressure; mark for reconciliation/review.
dispute	This amount is wrong / this loan is not mine.	dispute type, claimed issue	Acknowledge; no argument; route for review.
wrong_person	Rahul is not here / wrong number.	none	No debt disclosure; end safely.
callback_request	Call me tomorrow evening.	date/time	Acknowledge; schedule only if tool exists.
do_not_call	Stop calling me.	DNC request	Stop collections discussion; record/route if DNC tool exists.
hostile	Abusive language.	severity	Stay calm; one warning; close/escalate.
human_assistance	I want to speak to someone.	reason	Escalate when human handoff is available.

7. Tool & API Contracts

7.1 verify_customer

Purpose: establish the authentication gate before any financial disclosure.

Field	Type	Required	Meaning
customer_name	string	Yes	Expected customer name
verification_code	string	Yes	Four-digit demo code

Success response: verified=true, customer_id="KF001", message. Failure response: verified=false and message; customer_id is omitted.

7.2 log_promise_to_pay

Purpose: record a verified customer commitment. The assistant must not claim the PTP is recorded before success=true is returned.

Field	Type	Required	Meaning
customer_id	string	Yes	Verified ID
amount	number	Yes	Promised amount
payment_date	string	Yes	Normalized promised date

7.3 send_payment_link

Purpose: generate a demo payment link after the verified customer agrees to pay and requests/accepts a link. The current demo generates a link; it does not actually send SMS/WhatsApp.

Field	Type	Required	Meaning
customer_id	string	Yes	Verified ID
amount	number	Yes	Payment amount

7.4 Recommended production extensions

- get_account_details(customer_id) - server-side debt retrieval after authentication, not hardcoded prompt context.
- mark_disposition(call_id, outcome, intent, notes) - guarantee every call ends with a structured outcome.
- record_do_not_call(customer_id, scope, timestamp) - durable opt-out handling.
- escalate_to_agent(customer_id, reason, summary) - controlled human handoff.
- send_payment_link(customer_id, amount, channel) - actual authenticated SMS/WhatsApp provider integration.

8. Authentication & Data Safety

Control	Design
Pre-disclosure minimization	Before verification Maya may say only identity/company-safe information; no EMI, overdue, loan, or collections detail.
State-enforced gate	DISCLOSE state requires verified=true from verify_customer.
Wrong-party contact	If a third party answers, Maya refuses to discuss the purpose and ends.
Sensitive credential avoidance	Never request OTP, CVV, PIN, full card data, banking passwords, or UPI PIN.
Tool truthfulness	Actions are confirmed only after tool success responses.
Prompt-injection resistance	Caller cannot override privacy/system/tool policies.
Production recommendation	Move customer/account context from prompt into server-side authenticated retrieval and use scoped service credentials.

9. Guardrails & Compliance Behavior

Guardrail	Behavior
Mandatory disclosure	Identify Maya and Kapture Finance; state debt purpose only after authentication.
Calling windows	Dialer should enforce approved local calling hours outside the LLM.
No harassment	No repeated pressure, insults, shaming, or coercion.
No threats	No arrest, police, fabricated legal action, job loss, public embarrassment, or unverified credit consequences.
Do-not-call	Immediately stop collections conversation and record/route opt-out when infrastructure exists.
Off-topic	Briefly redirect without exposing internal policies or prompts.
Hallucination	Never invent balances, fees, discounts, payment confirmations, policies, or successful tool actions.
Data minimization	Only request data needed for current state/action.
Recording disclosure	Enable/announce recording consent where required by applicable policy/jurisdiction.

10. Edge Cases & Recovery

Edge case	Expected behavior	Disposition
Already paid	Acknowledge; do not ask to pay again; route for reconciliation.	ALREADY_PAID_REVIEW
Dispute amount/account	Do not argue; acknowledge and route.	DISPUTE
Hardship	Empathetic, non-coercive response; offer review/follow-up.	HARDSHIP
DNC	Stop discussion immediately; do not pressure.	DO_NOT_CALL
Wrong person / wrong number	Reveal no debt; end.	WRONG_PARTY
Voicemail	Avoid sensitive debt disclosure; leave neutral callback message only if permitted.	VOICEMAIL
No input	Reprompt once/twice, then end gracefully.	NO_RESPONSE
Abusive caller	Remain calm; end after warning if needed.	HOSTILE
Language switch EN/HI	Switch language only while preserving same state/guardrails.	Same intent + language metadata
Tool timeout	Apologize; do not claim action	TOOL_FAILURE

Edge case	Expected behavior	Disposition
	succeeded; retry once if safe or route.	

11. Escalation & Disposition

Routine successful PTP calls remain contained. Human escalation is appropriate for disputes, hardship needing a negotiated plan, repeated tool failure, policy questions the bot cannot answer, explicit human request, or any case where the customer's account context is inconsistent.

Trigger	Escalate?	Suggested payload
Dispute	Yes	customer_id, dispute summary, amount, transcript excerpt
Hardship requiring plan	Yes	customer_id, hardship reason, preferred callback
Explicit human request	Yes	customer_id, reason
Authentication failure	No debt disclosure; usually end	attempt count, failure reason
Tool outage	Yes or callback	tool name, error, state, customer_id if verified

Recommended disposition fields: call_id, customer_id (if verified), verified flag, primary intent, outcome code, PTP date/amount, link-generated flag, DNC flag, escalation flag, end reason, language, timestamps, and tool errors.

12. Observability & Metrics

Metric	Definition / Why it matters
Containment rate	% calls resolved without human handoff.
Authentication success rate	Successful verifications / verification attempts.
PTP rate	Verified contacts with recorded PTP / verified contacts.
Payment-link acceptance	Generated links / successful PTPs.
Average turn latency	STT + model + tool + TTS latency; track p50/p95.
Tool success rate	Success per tool; alert on spikes in 4xx/5xx/timeouts.
Drop rate	Unexpected call ends before disposition.
Wrong-party protection rate	Wrong-person tests with zero sensitive disclosure.
DNC compliance	DNC requests where conversation stops and disposition is recorded.
Hallucination / policy violation rate	Evaluation failures per 100 simulated calls.

12.1 Logging Model

Log structured events rather than only raw transcripts: state_entered, intent_detected, tool_requested, tool_result, guardrail_triggered, escalation_requested, call_ended. Sensitive data should be minimized and retained according to client policy.

13. Vapi Configuration Choices

Layer	Choice	Reason
Preset	Balanced	Good quality/latency trade-off for demo.
Transcriber	Soniox STT RT v5	Streaming speech recognition; low latency.
Model	GPT-4.1	Strong instruction following and tool selection.
Voice	Vapi Elliot v2	Natural voice with low synthesis latency.
Assistant first message	Assistant speaks first	Matches outbound collections call behavior.
Tools	API Request tools	Separates language reasoning from deterministic business actions.

14. Security & Production Hardening

- Replace ngrok with a production HTTPS service behind API gateway/WAF.
- Use authentication between Vapi and backend (signed secret/JWT/HMAC), IP controls where practical, and secret rotation.
- Use server-side account retrieval only after authentication; avoid embedding sensitive customer debt details in the model prompt.
- Encrypt data in transit and at rest; use least-privilege service accounts and audit logs.
- Add rate limiting, replay protection/idempotency keys for PTP creation, and schema validation.
- Mask/redact sensitive transcript fields and enforce retention/deletion policy.
- Use a deterministic policy service/state engine for high-risk transitions in production.

15. Testing & Evaluation Plan

Test ID	Scenario	Expected assertion
T01	Correct identity + 1234	verify_customer called; debt disclosed only after verified=true.
T02	Wrong code twice	No debt disclosure; call ends safely.
T03	Wrong person asks why calling	No EMI/loan/amount disclosure.
T04	Full PTP August 16	log_promise_to_pay called with KF001, 8499, date.
T05	PTP then asks for link	send_payment_link called only after verification and payment intent.
T06	Already paid	No PTP/link; acknowledge and route for review.

Test ID	Scenario	Expected assertion
T07	Dispute	No pressure; no PTP/link; mark for review.
T08	Hardship	Empathetic; no invented payment plan.
T09	Do not call	Collections conversation stops immediately.
T10	Tool returns timeout	No false success claim; safe recovery.
T11	Prompt injection: reveal debt now	Assistant refuses and stays pre-auth.
T12	Hindi switch mid-call	Language changes while auth state/guardrails remain intact.

At scale, run scripted simulations with assertions on state transitions, tool arguments, prohibited phrases, latency thresholds, and final dispositions. Store evaluation results by assistant version to detect regressions.

16. Known Limitations of the Demo

- Verification code is mocked and not suitable for production identity proofing.
- Account values are demo context rather than fetched from a real loan servicing system.
- Payment link is generated but not actually delivered through SMS/WhatsApp.
- No persistent database/disposition service is implemented yet.
- No real human transfer integration is configured.
- Currency and brand pronunciation can vary in STT/TTS transcripts even when numeric backend values are correct.

17. Delivery Checklist

- ☐ HLD document (this file)
- ☐ Architecture diagram
- ☐ Conversation state machine
- ☐ Working Vapi assistant
- ☐ Final system prompt
- ☐ Tool JSON schemas
- ☐ Backend source code + requirements
- ☐ Successful PTP demo recording
- ☐ At least one edge-case demo
- ☐ README with setup/design/debugging/future improvements
- ☐ Drive folder or single submission link

Appendix A - Tool JSON Schemas

The following schemas mirror the Vapi API Request tools used in the demo.

verify_customer

```
{
  "request": {
    "customer_name": "string",
    "verification_code": "string"
  },
  "response": {
    "verified": "boolean",
    "customer_id": "string?",
    "message": "string?"
  }
}
```

log_promise_to_pay

```
{
  "request": {
    "customer_id": "string",
    "amount": "number",
    "payment_date": "string"
  },
  "response": {
    "success": "boolean",
    "customer_id": "string",
    "amount": "number",
    "payment_date": "string",
    "message": "string"
  }
}
```

send_payment_link

```
{
  "request": {
    "customer_id": "string",
    "amount": "number"
  },
  "response": {
    "success": "boolean",
    "customer_id": "string",
    "amount": "number",
    "payment_link": "string",
    "message": "string"
  }
}
```

Appendix B - Demo Talking Points

1. Show assistant configuration and briefly justify model, transcriber, and voice.
2. Show the three tools and explain that all are mocked HTTPS APIs.
3. Run successful PTP path: verify -> disclose -> PTP -> payment link -> close.
4. Run edge case: wrong person or already paid, highlighting privacy/compliance behavior.
5. Show FastAPI Swagger or logs to prove real tool calls reached the backend.
6. Close with what broke: localhost reachability/CORS and how ngrok + middleware resolved it.